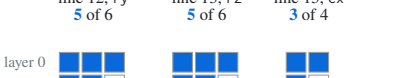
① srcML of the loop nest (lines 10–15)

```
<for>ᵉfor <control><expr><name>layer</name></expr><range>in <expr><call><name>range</name><argument_list>(<argument><expr><name>
    ↪ layersᶠ</name></expr></argument>)</argument_list></call></expr></range></control><block>:<block_content>
    <for>ᵉfor <control><expr><name>qubit</name></expr><range>in <expr><call><name>range</name><argument_list>(<argument><expr><name>
        ↪ nᶠ</name></expr></argument>)</argument_list></call></expr></range></control><block>:<block_content>
        <expr_stmt><expr><call><name><name>qc</name><operator>.</operator><name>ryᵇ</name></name><argument_list>(<argument>angle...
            ↪ </argument>, <argument><expr><name>qubitᶜ</name></expr></argument>)</argument_list></call></expr></expr_stmt>
        <expr_stmt><expr><call><name><name>qc</name><operator>.</operator><name>rzᵇ</name></name><argument_list>(<argument>angle...
            ↪ </argument>, <argument><expr><name>qubitᶜ</name></expr></argument>)</argument_list></call></expr></expr_stmt>
</block_content></block></for><for>ᵉfor <control><expr><name>qubit</name></expr><range>in <expr><call><name>range</name>
    ↪ <argument_list>(<argument><expr><name>nᶠ</name><operator>-</operator><literal type="number">1</literal></expr></argument>)
    ↪ </argument_list></call></expr></range></control><block>:<block_content>
        <expr_stmt><expr><call><name><name>qc</name><operator>.</operator><name>cxᵇ</name></name><argument_list>(<argument><expr>
            ↪ <name>qubitᶜ</name></expr></argument>, <argument><expr><name>qubitᶜ</name><operator>+</operator><literal type="number">1
            ↪ </literal></expr></argument>)</argument_list></call></expr></expr_stmt>
</block_content></block></for></block_content></block></for>
```

ᵉ `for` → unrolled ᶠ `range` → iteration count, from lines 4–5 ᵇ gate ᶜ qubit argument → a qubit per iteration *angle...* elided, kept symbolic

② Unrolled operations in the slice of q_0



Filled: in the wire-rule slice of q_0 (13 of 16 operations). The lines are the same in every iteration, but in the second layer the rotations of q_2 and its `cx` *are not dependences of q_0; a line-level slice keeps lines 12, 13 and 15 anyway.*